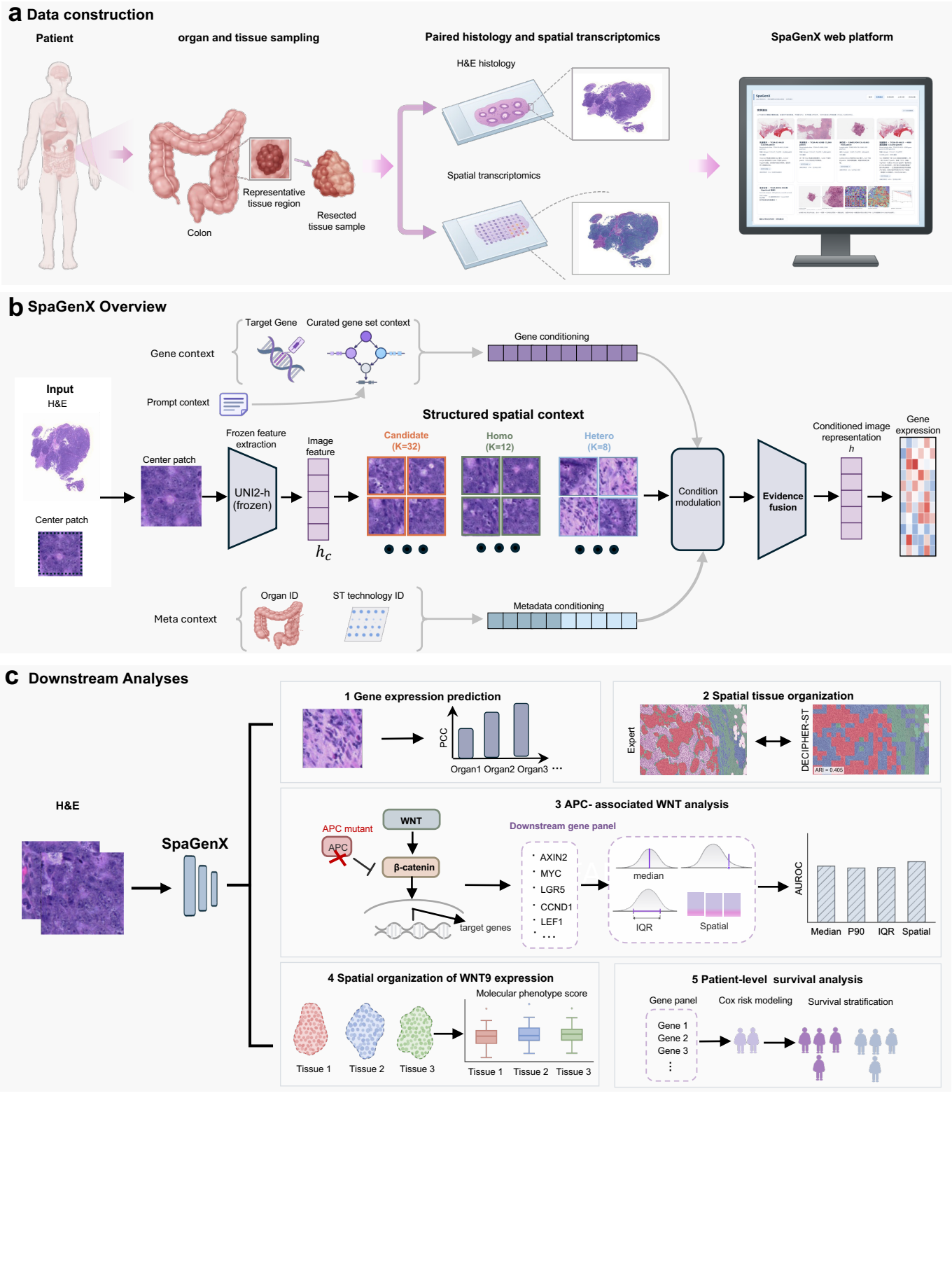

a Data construction
Patient
organ and tissue sampling
Paired histology and spatial transcriptomics
SpaGenX web platform
H&E histology
Spatial transcriptomics
Representative tissue region
Resected tissue sample
Colon
b SpaGenX Overview
Input
H&E
Center patch
Gene context
Target Gene
Curated gene set context
Gene conditioning
Prompt context
Structured spatial context
Center patch
Frozen feature extraction
UNI2-h (frozen)
Image feature
h_c
Candidate (K=32)
Homo (K=12)
Hetero (K=8)
Condition modulation
Evidence fusion
Conditioned image representation h
Gene expression
Meta context
Organ ID
ST technology ID
Metadata conditioning
c Downstream Analyses
H&E
SpaGenX
1 Gene expression prediction
PCC
Organ1 Organ2 Organ3 ⋯
2 Spatial tissue organization
Expert
DECIPHER-ST
ARI = 0.405
3 APC- associated WNT analysis
APC mutant
APC
WNT
β-catenin
target genes
Downstream gene panel
AXIN2
MYC
LGR5
CCND1
LEF1
median
IQR
Spatial
AUROC
Median P90 IQR Spatial
4 Spatial organization of WNT9 expression
Molecular phenotype score
Tissue 1 Tissue 2 Tissue 3
Tissue 1 Tissue 2 Tissue 3
5 Patient-level survival analysis
Gene panel
Cox risk modeling
Survival stratification
Gene 1
Gene 2
Gene 3
⋮